

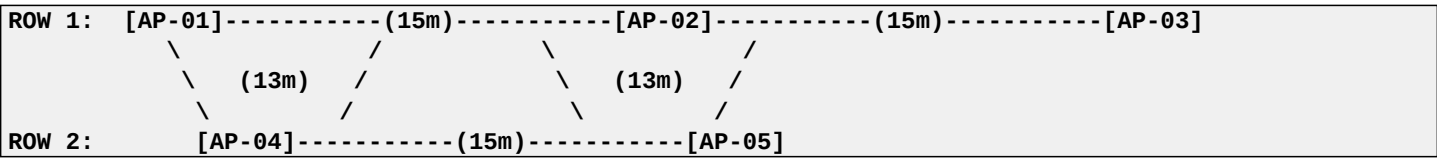
ULTIMATE INDOOR RF ARCHITECTURE & DESIGN GUIDE

1. THE TRIANGULAR HONEYCOMB CONCEPT

The Triangular Grid layout involves positioning Access Points (APs) at the vertices of equilateral triangles. This architecture is superior to square grids for Voice + Mobilization because it ensures that any 'weak spot' in the signal is equidistant from three different APs, providing a minimum of N+1 or even N+2 redundancy.

2. STAGGERED SPACING & 15-METER RULE

For Enterprise-grade Voice (VoWiFi), a -65 dBm signal at the cell edge is required. The 'Expert Rule' is a 15-meter horizontal spacing between APs. To achieve a perfect triangle, the vertical distance between rows must be 13 meters (calculated as $15\text{m} * 0.866$).



3. WALL ATTENUATION & dBm LOSS CALCULATIONS

RF signals are absorbed or reflected by physical barriers.

WALL ATTENUATION TABLE (5GHz Standard):

- Drywall / Office Partition: 3 dB to 4 dB
- Standard Glass: 2 dB to 3 dB
- Brick Wall: 8 dB to 10 dB
- Concrete Wall (Thick): 12 dB to 15 dB
- Metal / Elevator Shaft: 20 dB to 25 dB (Total Blockage)

EXPERT CALCULATION:

Loss = (Number of Walls * dB Loss per Wall). If you have two drywalls, the total loss is 8dB. This significantly reduces the 15-meter spacing rule.

4. IMPACT OF BARRIERS ON SPACING

If a concrete wall exists between two vertices of your triangle, the 15-meter rule must be reduced to 8-10 meters to compensate for the 12dB drop, ensuring the user still receives -65 dBm for fast mobilization.

5. MOBILIZATION STANDARDS (FAST ROAMING)

To support 'Fast Mobilization' for Voice, you must enable 802.11r (Fast Transition), 802.11k (Assisted Roaming), and 802.11v (Network Transition). These protocols allow the client to transition between APs in the triangle grid without dropping packets.